

Short-run Equilibrium in Retail: Navigating Strategies for Business Success

Case

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This case was written by Nour Wehbe (Montpellier Business School). It is intended to be used as the basis for class discussion rather than to illustrate either effective or ineffective handling of a management situation. The case was compiled from generalised experience.

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Case study

Managerial Context

Jayson, the manager of a retail store, is determined to enhance the short-term equilibrium and increase sales through a targeted workshop for his employees. The workshop starts with an in-depth exploration of key economic concepts, including cost functions, profitability thresholds, and partial short-term competitive equilibrium. The workshop involves detailed equilibrium analysis and interactive discussions on short-term market dynamics. Through these exercises, the aim is to provide the workshop participants with a thorough understanding of economic principles as they apply to real-world business situations. The ultimate goal is to empower Jayson's team with the knowledge and insights necessary to make informed decisions and strategically navigate the complexities of their retail business environment.

Scenario 1

In the first scenario, Jayson assumes that a firm producing phone cases for the Apple brand is characterized by the following total cost function :

$$C(y) = 3y^2 + y + 48$$

where y represents the quantity of goods produced, and we denote p as its unit market price.

Jayson aims to delve with his team into questions related to the efficiency and profitability of the production process. By delving into these aspects, Jayson and his team can gain a comprehensive understanding of the company's short-run profitability, make informed decisions about production levels, and respond effectively to market dynamics in the Apple phone case industry.

Scenario 2

In the second scenario, Jayson introduces a firm operating in a perfectly competitive market that produces fans. The firm's production function is given by :

$$y = 2x_1^{1/2}x_2^{1/3}$$

where y represents the level of output produced and x_1, x_2 the quantities of factors 1 and 2, respectively. We reason in the short-term with $\bar{x}_2 = 125$ and we assume that the prices of the production factors are respectively given by $w_1 = 300$ and $w_2 = 8$.

Jayson aims to delve into questions that allow for an understanding of the short-run equilibrium and surplus in this perfectly competitive market.

Scenario 3

In the third scenario, Jayson presents companies producing syringes for various uses (vaccines, insulin, etc.). Suppose that these companies are all identical and that this branch is characterized by the following total cost function :

$$C(y) = \frac{y^2}{2000} + 80$$

with y the number (in hundreds) of syringes produced by each company. The overall demand in this market is characterized by the following relationship :

$$X(p) = 52000 - 1000p$$

with p the price of the batch of a hundred syringes.

Jayson mentions for his team that, in this scenario, we reason on the short-run equilibrium in addition to the long-term market equilibrium.

Workshop's Purpose

Short-term equilibrium is a state in which the quantity supplied equals the quantity demanded in a market, leading to stability in prices and production levels. The purpose of this workshop is to provide Jayson and his team with a comprehensive understanding of short-term equilibrium analysis in economics. The goal is to understand the conditions under which a market or firm operates in the immediate future, typically within a brief time horizon.

References for further readings

Nicholson, W., & Snyder, C. (2008). Microeconomic theory : Basic principles and extensions. Mason, Ohio : Thomson/South-Western.

Shephard, Ronald W, (1981). Cost and Production Functions, SpringerVerlag Berlin Heidelberg.